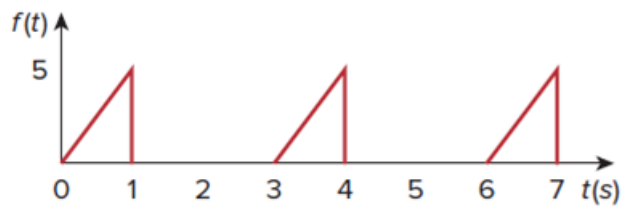


Problem

Determine the Laplace transform of the function in Fig. 15.27.

Figure 15.27:



Step-by-step solution

Step 1 of 4

Consider the following signal $f(t)$:

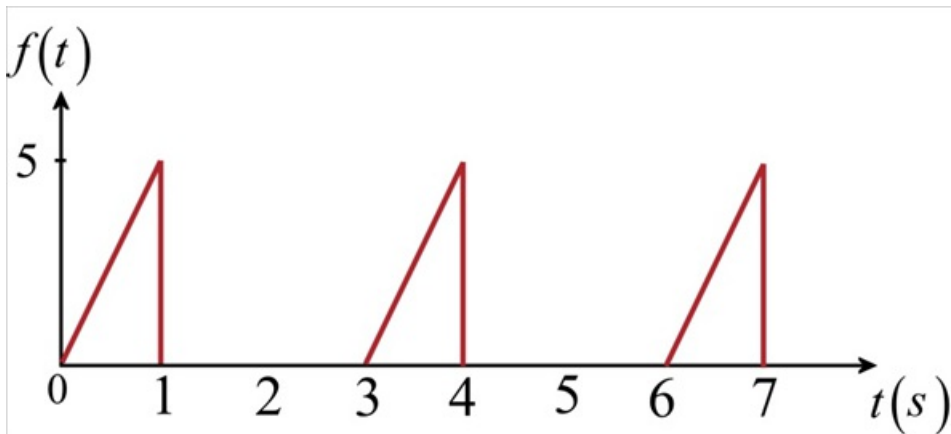


Figure 1

Step 2 of 4

Find the Laplace transform of the signal $f(t)$:

Clearly from Figure 1, $f(t)$ is a periodic signal of time-period 3 sec.

Therefore, Laplace transform of the signal $f(t)$ is $F(s) = \frac{F_1(s)}{1 - e^{-sT}}$.

Substitute 3 for T in this expression.

$$F(s) = \frac{F_1(s)}{1 - e^{-3s}} \dots\dots (1)$$

Where, $F_1(s)$ is the laplace transform of $f_1(t)$ which is $f(t)$ over its first period only.

Represent the signal $f_1(t)$ in mathematical form:

$$\begin{aligned} f_1(t) &= 5t[u(t) - u(t-1)] \\ &= 5tu(t) - 5tu(t-1) \\ &= 5tu(t) - 5(t-1+1)u(t-1) \\ &= 5tu(t) - 5(t-1)u(t-1) - 5u(t-1) \end{aligned}$$

Step 3 of 4

Calculate the Laplace transform of the signal $f_1(t)$.

Apply Laplace transform to the signal $f_1(t)$.

$$\begin{aligned} F_1(s) &= \mathcal{L}[5tu(t) - 5(t-1)u(t-1) - 5u(t-1)] \\ &= \mathcal{L}[5tu(t)] - \mathcal{L}[5(t-1)u(t-1)] - \mathcal{L}[5u(t-1)] \\ &= 5\mathcal{L}[tu(t)] - 5\mathcal{L}[(t-1)u(t-1)] - 5\mathcal{L}[u(t-1)] \\ &= 5\left(\frac{1}{s^2}\right) - 5\left(\frac{1}{s^2}e^{-s}\right) - 5\left(\frac{1}{s}e^{-s}\right) \end{aligned}$$

$$F_1(s) = \frac{5}{s^2}(1 - e^{-s} - se^{-s})$$

Therefore, the Laplace transform of the signal $f_1(t)$ is $\frac{5}{s^2}(1 - e^{-s} - se^{-s})$.

Step 4 of 4

Substitute $\frac{5}{s^2}(1 - e^{-s} - se^{-s})$ for $F_1(s)$ in equation 1.

$$\begin{aligned} F(s) &= \frac{1}{1 - e^{-sT}} \frac{5}{s^2}(1 - e^{-s} - se^{-s}) \\ &= 5 \frac{(1 - e^{-s} - se^{-s})}{s^2(1 - e^{-3s})} \end{aligned}$$

Therefore, the Laplace transform of the signal $f(t)$ is $\boxed{5 \frac{(1 - e^{-s} - se^{-s})}{s^2(1 - e^{-3s})}}$.